

Graph-Based Optimization for High-Density LEO Constellation Design

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ABSTRACT

Graph-based optimization for high-density LEO constellation design focuses on reducing the risk of collision events in the congested low Earth orbit (LEO) region through strategic orbital selection. A key approach involves developing a graph-theoretic method for planning high-density LEO constellations that minimize the probability of collisions with existing LEO space objects. This approach is extended for the constellation use case of LEO Space Traffic Management (STM), namely, maximizing space-to-space LEO observations. A promising method for achieving this optimization is simulated annealing optimization, known for handling large scale problems such as this. This work aims to enhance the safety and sustainability of space operations in the increasingly crowded LEO environment.

1. INTRODUCTION

The rapid increase in satellite deployments, particularly in low Earth orbit (LEO) such as Starlink, OneWeb, and China's Thousand Sails, has led to unprecedented congestion and heightened likelihood of conjunction events. With the expansion of mega-constellations, ensuring safe and sustainable space operations has become a critical challenge. Effective space traffic management (STM) strategies must be developed to mitigate collision risks and optimize orbital utilization. As commercial, governmental, and defense stakeholders continue to launch satellites, the urgency for structured, scalable processes to manage the increasingly congested LEO environment grows ever more critical. This paper explores advanced methods for planning high-density constellations within already highly congested LEO regions, focusing on minimizing conjunction risk and maximizing space-to-space LEO observations as well as Earth land observations.

We propose a fundamental approach to this problem in the development of a graph-theoretic framework that models satellites and their potential conjunctions with LEO objects as nodes and weighted edges. This structure enables the application of optimization techniques to reduce collision risk while maximizing space-to-space LEO observations. It also supports efficient evaluation of constellation configurations in large-scale, high-density scenarios. This method will be particularly useful for the design of high-density or mega-constellations on the order of hundreds to thousands of satellites, ensuring the safety of the entire LEO environment.

A promising technique for general large-scale graph optimization is simulated annealing. Simulated annealing is an optimization algorithm that provides a scalable method to optimize high-dimensional problems, as is this problem of constellation design. We formulate our graph by modeling potential constellation satellites as nodes and interactions between those nodes and the LEO object population as weighted edges. The simulated annealing algorithm then identifies satellite distributions that optimize our mission objectives – namely, minimizing collision risk while also optimizing one or more additional metrics such as LEO coverage. Since the positional uncertainty of LEO objects is highly dependent on their altitude, thus impacting the probability of conjunction, the implemented probability metric models this uncertainty.

Extensive prior work has highlighted both the scale and implications of deploying large LEO constellations. Zhang et al. [1] provide a comprehensive overview of current and planned mega-constellations, detailing their operational characteristics and emphasizing the potential for orbital congestion and increased conjunction risk. Further, [2] discusses the safety implications of these large-scale deployments, outlining the need for proactive risk assessment and mitigation strategies to prevent collisions in densely populated orbital regimes. Lifson et al. [3] reviews

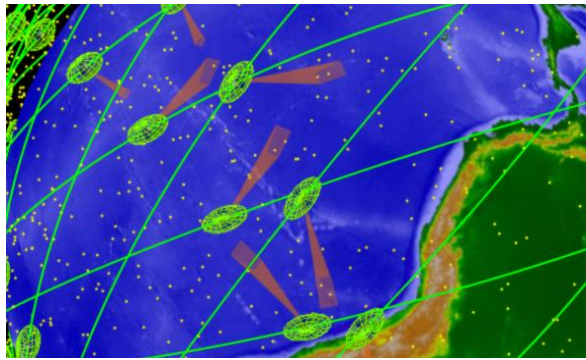
historical approaches to satellite orbit packing and offers criteria for the efficient arrangement of satellites, which can inform the design of high-density constellations by establishing guidelines for safe inter-satellite spacing. Complementing these efforts, Arnas [4] leverages the concept of two-dimensional Lattice Flower Constellation (2D LFC), providing a quantitative framework to define ‘safe’ satellite distributions and offering a foundation for assessing orbital slot allocations. We will use this paper as an input for our graph as a starting point of ‘safe’ satellite distributions. From this, we will build our graph and optimally select the satellites for our constellation that meet our desired objectives.

Previous work by Stevenson [5] demonstrates the utility of representing satellite interactions as networks, enabling the use of graph-based algorithms for assessing conjunction risk. While our work also assesses conjunction risk, it deviates from the work by Stevenson in that our aim is to use conjunction risk information (as well as coverage information) for constellation design. In our work, potential constellation satellites are modeled as nodes and potential collisions between node pairs and LEO objects as weighted edges. This graph definition and approach of using simulated annealing to optimize this constellation design provides novel and scalable solutions for arranging high-density constellations under complex operational constraints.

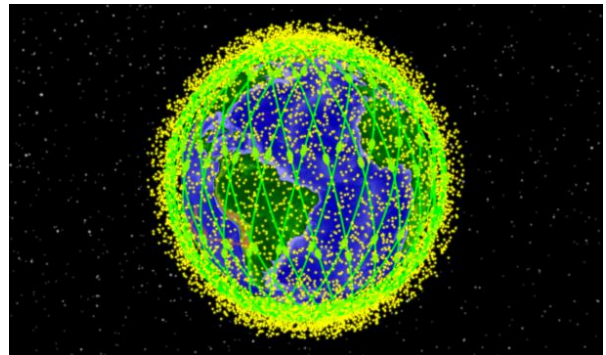
2. BACKGROUND

2.1 Overview

The primary objective of this work is to develop a method for designing LEO mega-constellations in already dense regions that minimize collision risk, both within the constellation itself and with the broader LEO space object population. This involves selecting optimal satellites that have minimum risk of colliding with one another as well as all LEO objects. This framework can also be extended to include additional mission objectives. In this paper, we focus on two examples: (1) LEO coverage and (2) Earth land coverage. Fig. 1a illustrates the LEO coverage problem, where constellation satellites and their position uncertainty ellipsoids are separated from other constellation members as well as from existing LEO objects. Fig. 1b shows the current LEO space object population alongside the orbits of a proposed mega-constellation. As discussed in Section 3, this visualization naturally parallels the graph-theoretic formulation of the problem.



(a) LEO STM constellation



(b) LEO population with STM constellation

Fig. 1. Constellation design to minimize collision risk while maximizing coverage of LEO. (a) Constellation satellites with position uncertainty (green ellipsoids) and sensors observing LEO object population (yellow objects). (b) Example architecture with the entire LEO object population.

Constellations under design are initialized with a shell of satellites that are maximally packed while minimizing inter-constellation collision risk. From this starting point, additional objectives can be incorporated into the design process. Graph theory provides the mathematical framework for constructing mega-constellations that balance these objectives. The overarching goal of these strategies is to improve the safety and long-term sustainability of space operations in an increasingly congested orbital environment. The following subsection introduces the current distribution of objects in LEO.

2.2 Density of LEO

The work in this paper is motivated by the rapid increase in the LEO object population due the deployment of mega-constellations. The trend of mega-constellations began with the first Starlink launches in 2019 [6]. Other companies

are in the process of launching proliferated satellite constellations into the LEO regime, including OneWeb, Amazon (Kuiper), and the Chinese Thousand Sails (Qianfan) constellation. Fig. 2 shows the inclination vs altitude distribution of many of these constellations as it stands for July 2025 as reported in the unclassified Space-Track catalog [7].

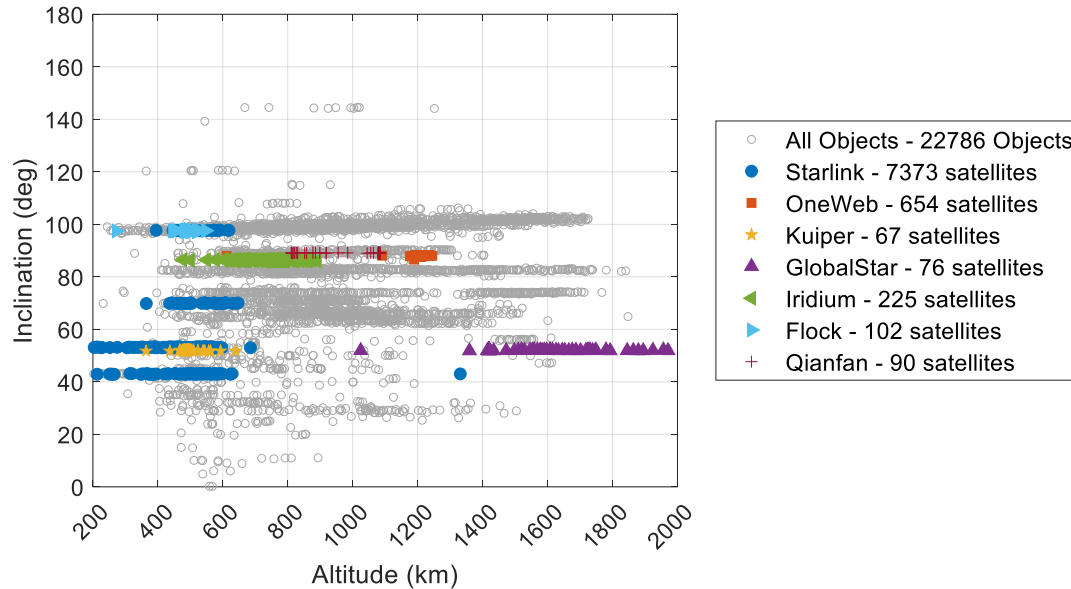


Fig. 2. Inclination vs altitude distribution of various mega-constellations vs the entire LEO object catalog.

LEO is already densely populated due to the large number of debris objects generated from two Anti-Satellite (ASAT) tests on the Chinese satellite Fengyun-1C in 2007 [8] and the Russian satellite Cosmos-1408 in 2021 [9]. The Cosmos-Iridium satellite collision in 2009 [10] has also contributed to the large number of debris objects in LEO. As of July 2025, the LEO object population consists of 8,851 debris objects, 11,999 payloads, 928 rocket bodies, and 1,008 other objects.

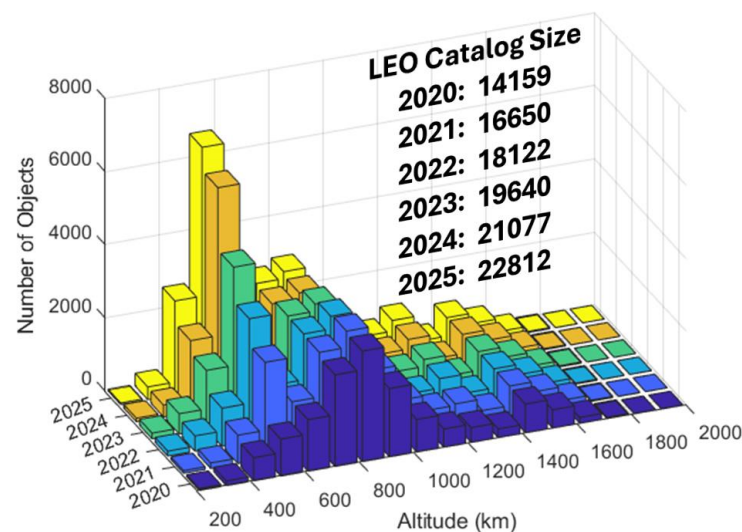


Fig. 3. History of altitude distribution of LEO object population.

Disproportional growth has occurred at specific altitude bands. Fig. 3 shows the distribution of LEO objects by altitude for the years 2020 – 2025 [7]. The largest growth has occurred between altitudes of 550 and 600 km, which corresponds to the altitude of many of the Starlink satellites, as well as Flock, Iridium, and Kuiper satellites.

If future planned LEO satellite deployments are realized, the number of objects in LEO could increase by as much as 100,000 [11]. LEO satellite architectures should consider the safety of the entire LEO regime as well as the burden placed on organizations, such as the space surveillance network (SSN) that are tasked with tracking all artificial objects in Earth orbit [12]. A method to plan for the safe deployment of future mega-constellation is useful for the safety of all objects in LEO. Section 3 will present an approach for planning reduced risk LEO constellations.

2.3 Graph Theory

Graph theory has gained traction in optimization for a variety of reasons. Graphs are an intuitive way of visualizing large-scale problems in terms of nodes and edges. Nodes are defined as the entities or decision variables in a system (e.g., satellites, cities, processors, or states in a search space), and edges are defined as the relationships or interactions between those entities (e.g., communication links, transportation routes, or probabilities of transition). Graph theory has been used for applications as varied as modelling the structure of bio-molecular networks [13], energy routing optimization [14], and satellite constellation design [15].

In our problem, nodes represent potential satellites for our constellation and edges represent a combination of collision risk and a coverage metric. In Fig. 4, a graph representation is visualized. The gray nodes and edges represent the full graph, and the green nodes represent the selected subgraph that is the solution. The solution is the subset of nodes that contain the maximized sum of edge weights between selected nodes. Each node contains edges that connect to each node in the graph. Fig. 4 only contains a total of only 40 nodes, yet it already contains 780 edges and over 650,000 possible solutions (solutions of size 5 nodes). One can see how these problems quickly become intractable to solve using brute force. Fortunately, graphs are convenient ways to establish problems for optimization algorithms to solve.

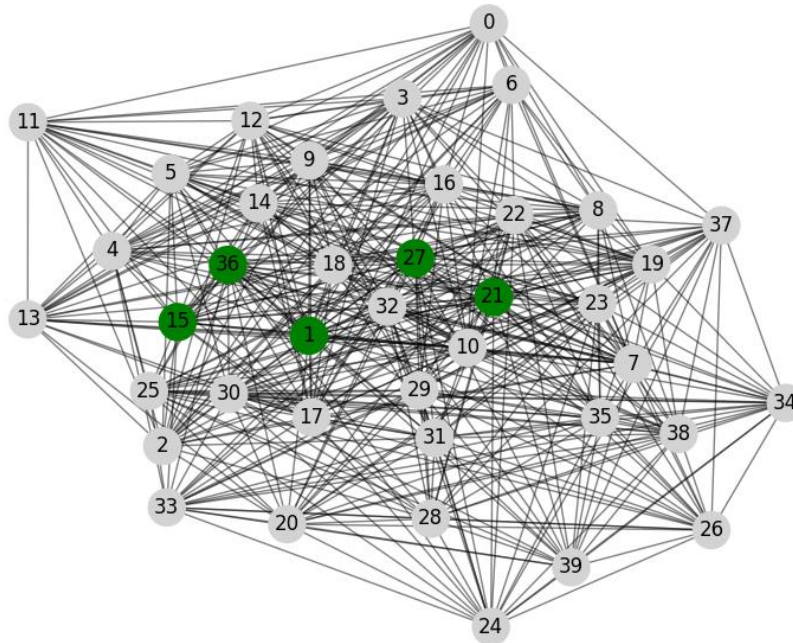


Fig. 4. Graph example with 40 total nodes, with 5 selected (green). Each edge connects a node to each of the other nodes.

3. APPROACH

3.1 Formulating the Graph

When constructing graphs with the intent of optimization, it is vital to construct these graphs carefully, with the optimization process in mind. In the case of this paper, we propose a method of designing high-density or mega-constellations on the order of hundreds to thousands within already dense regions in LEO. This constellation will be constructed with the goal of minimizing collision risks while also maximizing space-to-space LEO coverage.

It should be noted that we define “filtered” LEO objects as LEO objects that have undergone both an apogee-perigee filter and a large screening volume filter as described in Section 3.1.1.

Define the following notation in Table 1:

Table 1. Definitions of notation

Notation	Definition
v_n	Vertex n that represents satellite n
P_{cn}	Probability of collision between satellite n and all filtered LEO objects
a_n	Percent of LEO access from satellite n
L_n	Total number of filtered LEO objects relative to v_n
$P_{c_{ln}}$	Probability of collision between v_n and a single LEO object l
k	Number of satellites in final constellation
$G(V, E)$	Graph with V and E denoting vertices/nodes and edges, respectively
N	Total possible satellites for constellation
x, y	Scaling factors for P_c and coverage
obj_l	Single LEO object l

Our particular graph set-up will rely on k -node subgraphs, where k = the desired number of satellites in our constellation. Let $G(V, E)$ represent our graph with V and E denoting vertices/nodes and edges, respectively. Then the input nodes are $\{v_n: v_n \in V, n = 1, \dots, N\}$ where N = total possible satellites. This problem is $\binom{N}{k} = \frac{N!}{k!(N-k)!}$ complex, thus a brute-force approach to solving this is infeasible.

We define the edge weight between nodes v_1, v_2 as the probability that neither node (representing satellites) collides with an object in LEO as well as the average percent of LEO objects accessed by both nodes.

Specifically, define the edge weight as the following:

$$w(v_1, v_2) = x(1 - P_{c1})(1 - P_{c2}) + y(avg(l_{obs})), \quad (1)$$

where $avg(l_{obs})$ is the average LEO coverage between the nodes, P_{cn} is the probability of collision between node v_n and the entirety of filtered LEO objects, and x and y are scaling factors that can be tuned. Details of the coverage computation are covered in Section 4.2.

The total P_c for a whole constellation is computed by the following (and holds since each P_{cn} are statistically independent):

$$P_c = 1 - \prod_n (1 - P_{cn}). \quad (2)$$

Consequently, $P_{cn} \leq P_c \forall n \in constellation$. Details of the probability of collision will be defined in more detail in the following subsection.

To visualize this graph construction, consider two nodes, v_1, v_2 , with edge weight $w(v_1, v_2)$ in Fig. 5. Each of the vertices (representing satellites) are compared against the filtered LEO objects for each node’s probability of collision with these filtered LEO objects.

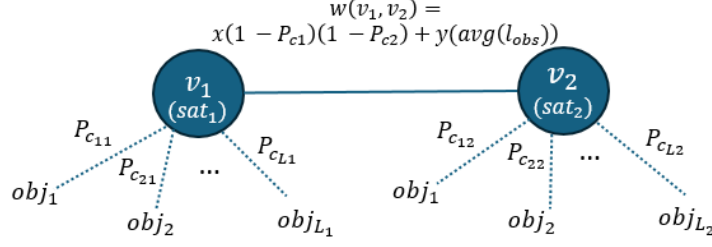


Fig. 5. Visual description of graph with two vertices, v_1, v_2 (representing input satellites 1 and 2) and the edge weight between them.

In this example,

$$\begin{aligned} P_{c1} &= 1 - \prod_l (1 - P_{c_{l1}}), \forall l \in L_1, \\ P_{c2} &= 1 - \prod_l (1 - P_{c_{l2}}), \forall l \in L_2, \\ avg(l_{obs}) &= \frac{(a_1 + a_2)}{2}. \end{aligned} \quad (3)$$

The objective of constructing this graph is to select k nodes that maximize the sum of weighted edges, thus minimizing the collision risk of the constellation with LEO objects and maximizing LEO coverage. We define our problem mathematically as the following:

$$\begin{aligned} \max_{v_i \in V} \sum_{n=1}^N x_n \sum_{m=1}^N w(v_n, v_m), \\ s. t. \sum_{n=1}^N x_n = k, \end{aligned} \quad (4)$$

where $x_n \in \{0, 1\}$ indicates whether node $v_n \in V$ was not selected or was selected, respectively (ensuring the desired number of satellites in the constellation is obtained). This equation is visualized by the graph in Fig. 6, for a graph with 4 nodes and 6 edges.

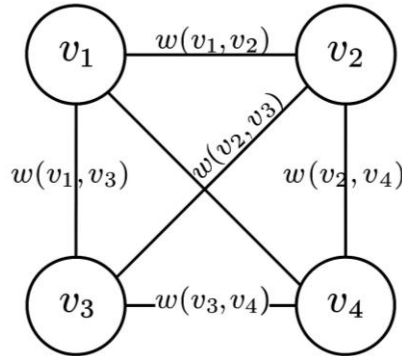


Fig. 6. Equation (4) visualized as graph where the goal is to choose the vertices that maximize the sum of w .

It should be noted that a part of this equation that is not shown is the conjunction risk between input satellite nodes. To account for this, the possible satellite nodes that are input are strategically placed in shells corresponding to [4]

and are thus guaranteed to not collide. The collision risk between these input satellite nodes were calculated through our collision risk assessment process and were verified to not collide with one another. The purpose of this was to decrease our search space, thus reducing computation to just the conjunction risk between each $v_n \in V$ and all filtered LEO objects.

3.1.1 Conjunction Screening

In this work, a standard conjunction screening process is utilized, with the use of collision probability (P_c) as the risk metric in the constellation design process. The conjunction screening process is shown in Fig. 7. We utilize this process to determine the P_c value between each member of the proposed constellations against all LEO objects, debris and non-debris.

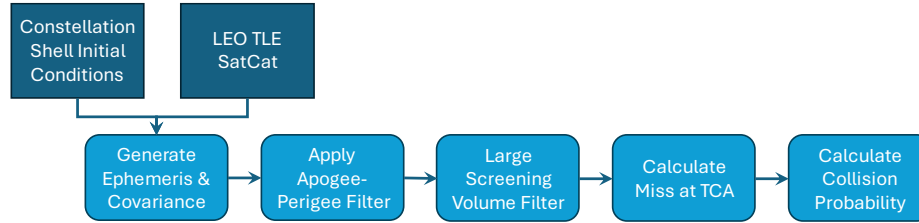


Fig. 7. Conjunction screening process used to assess collision risk for constellation members against LEO object catalog.

Two screening filters are used in this process, an apogee-perigee filter and a large screening volume filter. The apogee-perigee filter is a common filter used in conjunction screening processes. It allows for the removal of 1v1 object pairs that have no chance of colliding due to the separation of their orbits. Refer to [16] for a complete description of the filter. Next, a large screening volume filter is applied to reduce the number of close approach calculations. This filter removes all conjunctions that are outside a large sphere and will not pose a collision threat. This calculation is performed at common ephemeris time steps. The volume is chosen to guarantee that the objects will always remain at safe distances between time steps.

We use Chan's 2D analytic P_c equation for use in the P_c calculation [17]. The formula is,

$$P_c = e^{-\frac{v}{2}} \left(1 - e^{-\frac{u}{2}} \right), \quad (5)$$

$$\text{with } u = \frac{\rho^2}{\sigma_x \sigma_z} \text{ and } v = \frac{x^2}{\sigma_x^2} + \frac{z^2}{\sigma_z^2},$$

where ρ is the radius of the circular cross section representing the size of both objects. The remaining parameters reside in the conjunction plane that is orthogonal to the relative velocity. Combined position uncertainty is represented by σ_x and σ_z . The relative position of the objects is given by x and z .

This method was chosen due to its speed and ease of implementation. All close approaches in this work are assumed to have high relative velocity, hence the choice of a 2D P_c calculation method. This is an appropriate choice for this constellation design process since high relative velocity conjunctions represent the vast majority of events in LEO. The screening process and risk calculation that were implemented herein could be modified in practice to utilize 3D P_c calculations such as [18] or other screening filters.

3.2 Optimizing the Graph

As mentioned previously, a key motivating factor in constructing our problem as a graph is to conveniently apply optimization algorithms to this. After evaluating multiple algorithms to maximize this graph, simulated annealing showed the most promise. When applying simulated annealing to maximize our graph, the goal is to identify a subset of k nodes whose internal edges produce the highest total weight. This solves the problem of choosing satellites for a constellation of size k that minimizes the probability of collision and maximizes LEO coverage. The

simulated annealing algorithm begins with a random selection of k nodes and evaluates the sum of the connecting edge weights.

At each step, one node in the subset is exchanged with one outside of it, and the change in total weight is assessed. Moves that increase the objective are accepted, while moves that decrease it may still be accepted with a probability that diminishes as the temperature parameter cools. This mechanism allows the algorithm to explore a broader solution space and avoid becoming trapped in local maxima during the early stages of the search.

As the temperature declines, the algorithm becomes increasingly selective, focusing on moves that improve the total edge weight. Through this balance of exploration and exploitation, simulated annealing converges toward a configuration of k nodes that yields a near-optimal total internal edge weight.

This process can be visualized in Fig. 8. The left figure shows the initial weighted graph, where node connections are colored by edge weight (darker = higher weight). The figure on the right demonstrates the progression of the total internal edge weight over iterations, showing how the algorithm explores different node subsets and gradually converges toward a higher-weight solution.

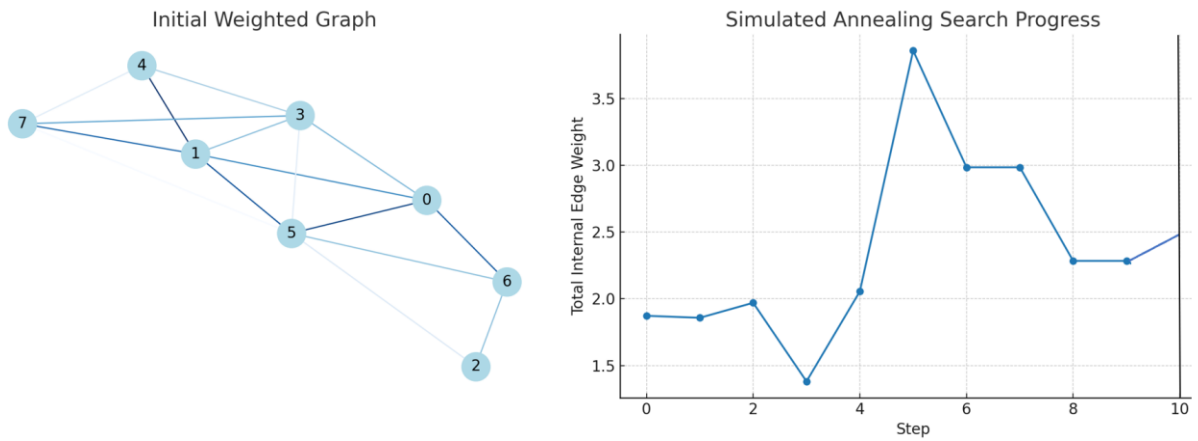


Fig. 8. Simulated annealing search process

4. SIMULATION SETUP

This section describes the simulation inputs and outputs for construction of the graph, as well as relevant parameters that are considered for our results.

4.1 Inputs

This problem is formulated as a “fleet vs all” problem, where a set of candidate satellites for the constellation are input and are each assessed against a set of LEO objects. The catalog of LEO objects is queried from Space-Track.org [7] as a TLE list of objects with apogee less than 2,000 km and eccentricity less than 1.0. The resulting catalog is composed of over 20,000 objects.

The “fleet” of potential satellites in the constellation are defined by 5 orbital shells, where satellites within a shell share the same inclination and altitude. Orbital parameters for each shell are defined as optimally packed 2-D lattice flower constellations, as defined in [4]. By definition, the constituent satellites in each shell are free of self-conjunctions and maintain at least 1 degree angular spacing (as seen from the Earth center) at all times between all other satellites within a shell. By design, these input satellites have zero probability of collision (P_c). These have been additionally verified to have $P_c = 0$ using the same metrics the “fleet vs all” P_c were calculated. The goal of our process is to optimally select satellites from all the various shells to construct a constellation that meets our objectives.

These shells are equivalently defined with circular Walker-Delta parameters, shown in Table 2.

Table 2. Orbital configuration of potential input satellites.

	Altitude (km)	Inclination (deg)	Number of Satellites	Number of orbital planes
Shell 1	400	15	1376	16
Shell 2	475	60	1722	246
Shell 3	550	30	1656	184
Shell 4	625	45	1869	267
Shell 5	700	75	1414	101

The selected inclinations correspond to the optimally packed results from [4], while the altitudes were selected to cover a range of high-density regions in the catalog. It is important to note that in the general method described in this paper, the selection of potential input satellites is arbitrary, but in this case were selected such that they are guaranteed to be free from self-conjunctions within shells and between shells. The goal of this was to reduce the computation time of also calculating P_c between satellites in the constellation. .

4.2 Mission Performance

To incorporate mission-specific parameters into the graph construction, performance is assessed for every potential input satellite. Two representative missions are considered for performance inputs: observation and coverage of the LEO catalog, and Earth land coverage. The potential input satellites are defined in Table 2. In each mission, the fleet represents a proliferated network of observers in LEO.

For observation and coverage of the LEO catalog, each satellite in the fleet is assessed for coverage, or access, of the debris and non-debris LEO catalog. At every time step in the three-day simulation, each satellite in the fleet determines whether it has access to every object in the catalog. Access is determined with a set of constraints as shown in Table 3.

Table 3. Space-to-space coverage constraints.

Constraint	Definition
Solar Exclusion	A viewing exclusion zone around the Sun. Access is possible if the angle between the viewer-Sun vector and viewer-target vector is greater than 30°
Earth Limb Exclusion	A viewing exclusion zone around the Earth. Access is possible if the target grazing angle (angle of the target's position above the horizon, relative to the viewer) is greater than 5°
Line of Sight	Access is possible if the viewer has LOS to the target
Target Sunlit	Access is possible if the target is sunlit
Target SNR	Access is possible if the target SNR is greater than the 6.0 SNR is a function of the target size, range, velocity, and solar phase angle relative to the viewer.

For the large, proliferated architectures under consideration, a low-cost visible spectrum payload was selected for evaluation and every satellite in the fleet is assumed to carry the same payload. The sensor has aperture diameter of 0.034 m, 0.075 m effective focal length, and cut-on and cut-off wavelengths of 0.4 μm and 1.2 μm . Space-to-space performance of this sensor is modeled with an internal tool – an SNR contour plot is shown in Fig. 9, as a function of target range and target velocity [19].

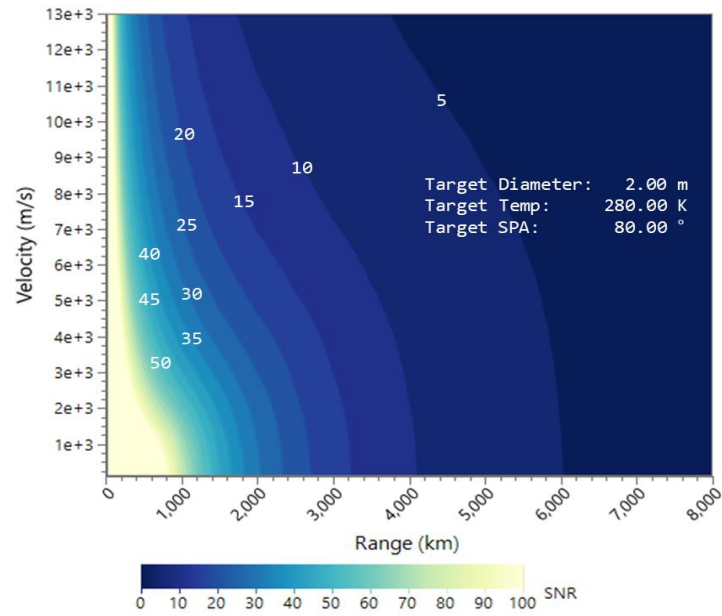


Fig. 9. Visible spectrum payload SNR by target range and velocity.

For Earth land coverage, each satellite in the fleet is assessed for coverage of an array of points distributed uniformly across inhabited landmass, as shown in Fig. 10. At every time step in the simulation, each satellite in the fleet determines if it has access to every point in the target array. Access is determined with the constraints shown in Table 4. Coverage in this problem does not consider radiometric detectability; access is purely a result of geometric constraints and lighting conditions.

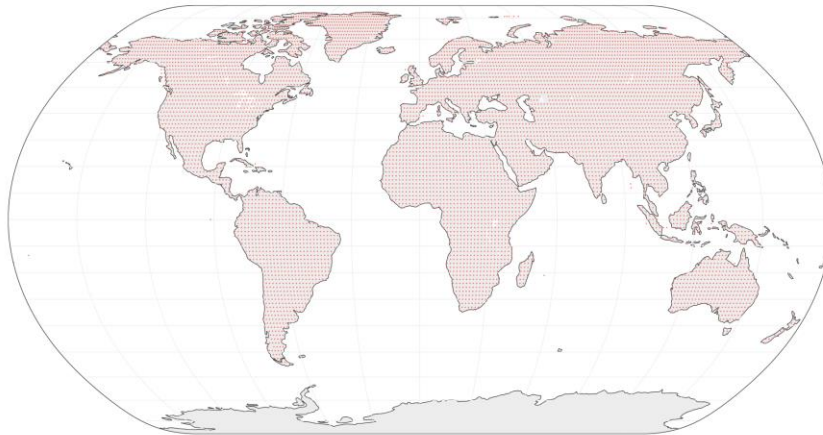


Fig. 10. Earth land coverage target points.

Table 4. Earth land coverage access constraints.

Constraint	Definition
Line of Sight	Access is possible if the viewer has LOS to the target
Field of Regard	Access is possible if the target is within the viewer's Field of Regard (FOR), defined as nadir-centered cone with half-angle of 20°
Ground Elevation	Access is possible if the viewer's angle above the horizon relative to the target is greater than 10°
Target Sunlit	Access is possible if the target is sunlit

After access is computed, various coverage-related metrics are calculated for every fleet-catalog pair including access duration (as a percent of the full simulation time), mean time to access (MTTA), minimum gap time, average gap time, and maximum gap time. For each member of the fleet these values are averaged over the entire catalog, resulting in aggregate measures of performance for each fleet satellite assessed against the full catalog. These per-satellite results are used in the construction of the graph.

4.3 Conjunction Risk Assessment

With the fleet and catalog inputs as defined in Section 4.1, the conjunction screening is performed for all fleet-catalog pairs for the three-day simulation time. The general risk threshold in the conjunction screening is set to 10 km. To compute the P_c between a fleet-catalog pair, the covariance must be defined and propagated for all objects in the scenario. For simplicity, the same spherical positional covariance ellipse is defined for every object in the fleet and catalog. In the cartesian Geocentric Reference Frame (GCRF), the positional covariance ellipse is defined as the identity matrix $I_3 \times 10,000$. Full details of the conjunction screening and risk analysis are covered in Section 3.1.1.

5. RESULTS

This section is primarily focused on results related to LEO coverage. However, one of the strengths of this method is its flexibility—it is not limited to a single mission type. To illustrate this point, we also include an example of Earth land coverage. While the drivers and dynamics of Earth land coverage differ significantly from those in LEO, the same underlying framework applies, highlighting the adaptability of the approach across a wide range of orbital environments.

5.1 Space Situational Awareness: LEO Coverage

To understand the performance of different constellation sizes, the simulated annealing algorithm was run to generate constellation sizes of 100, 500, and 1,000 satellites. Constellations were formed from each individual orbit shell listed in Table 2, as well as constellations that combine satellites from each shell.

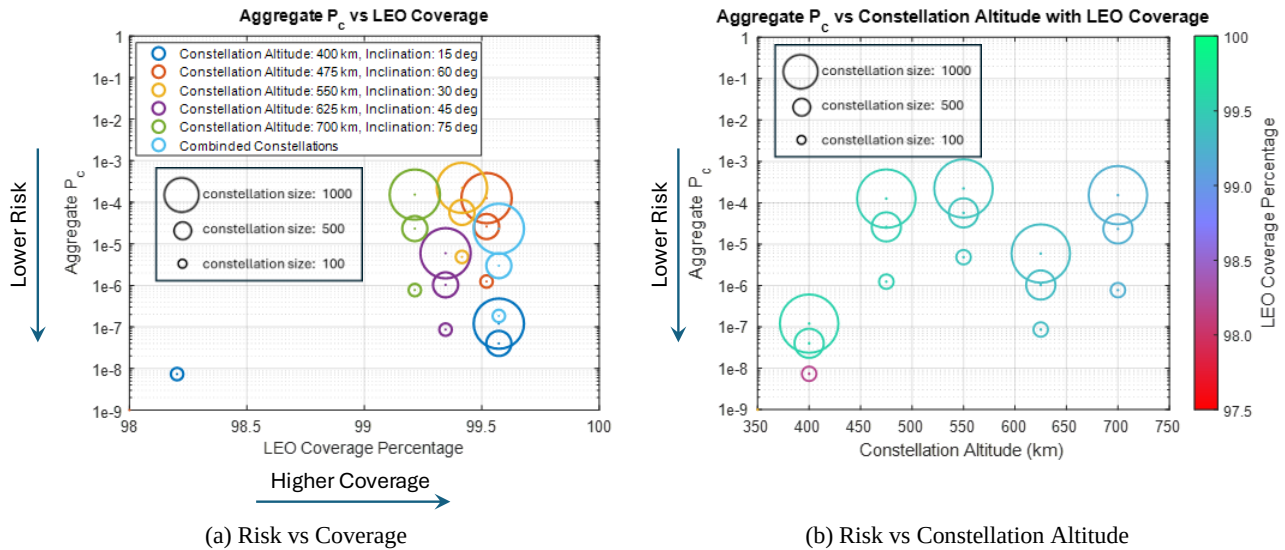


Fig. 11. Algorithm performance for various constellation sizes and altitudes for the LEO coverage mission.

The risk and coverage results of the constellations are displayed in Fig. 11, with constellation size designated by the size of the plot marker. Best performing architectures have low aggregate P_c and high coverage – in Fig. 11a these constellation results lie in the lower right portion of the plot. For the number of satellites analyzed, coverage is high for all constellations, so the aggregate P_c is the distinguishing metric. Given that the constellation at 400 km has the lowest risk, that could be a favorable orbit shell for a LEO STM architecture, with a constellation size of 100 satellites providing good coverage. The low risk of constellations at 400 km altitude is further shown in Fig. 11b, that displays aggregate P_c vs constellation altitude with coverage indicated through color. An order of magnitude reduction in risk is observed as compared to the other altitudes. The constellation of 100 satellites at 625 km has the

next lowest aggregate P_c , while also having a higher coverage percentage. These results are one piece of a constellation design process, with additional factors such as architecture cost, orbit maintenance, and end-of-life disposal also affecting the choice of the constellation orbits.

The results shown in Fig. 12 demonstrate that the algorithm is achieving the stated goal of reducing collision risk through the optimal placement of satellites. These plots display the aggregate P_c for each satellite within a constellation. The distribution of P_c values from the optimized constellations show reduced collision risk for constellation sizes of 100, 500, and 1000 satellites as compared to a random draw from the orbit shell. Results are shown for two orbit shells, (a) 550 km altitude, and (b) 625 km altitude.

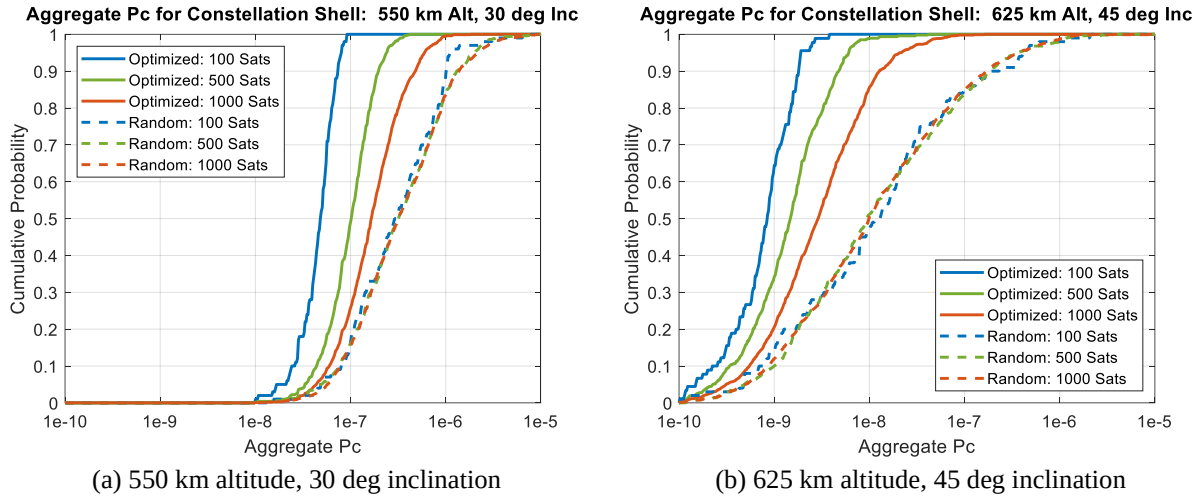


Fig. 12. Aggregate P_c as cumulative distributions for various constellation sizes. Optimized constellations compared to random draw from orbit shell. Data for two constellation orbit shells.

To further illustrate the reduction in risk achieved by the algorithm, tabular results of the aggregate P_c are presented in Table 5. The aggregate P_c was reduced by an order of magnitude or more for 14 of the 18 constellations by using the simulated annealing algorithm.

Table 5. Aggregate P_c for various constellation sizes. Optimized constellations compared to random draw from orbit shell.

Constellation Orbit	Constellation Size	Aggregate P_c Optimized Constellation	Aggregate P_c Random Constellation	Aggregate P_c (Order of Magnitude) Optimized vs Random
Altitude: 400 km Inclination: 15 deg	100	7.34E-09	4.60E-07	1.8
	500	4.01E-08	9.66E-06	2.4
	1000	1.20E-07	9.71E-06	1.9
Altitude: 475 km Inclination: 60 deg	100	1.24E-06	8.28E-05	1.8
	500	2.61E-05	2.33E-04	1.0
	1000	1.26E-04	5.76E-04	0.7
Altitude: 550 km Inclination: 30 deg	100	4.84E-06	7.99E-05	1.2
	500	5.69E-05	2.88E-04	0.7
	1000	2.22E-04	7.05E-04	0.5
Altitude: 625 km Inclination: 45 deg	100	8.63E-08	5.56E-06	1.8
	500	1.03E-06	4.27E-05	1.6
	1000	5.93E-06	8.44E-05	1.2
Altitude: 700 km Inclination: 75 deg	100	7.72E-07	4.16E-05	1.7
	500	2.33E-05	2.20E-04	1.0
	1000	1.53E-04	4.35E-04	0.5

Combined Constellation	100	1.84E-07	3.58E-05	2.3
	500	2.98E-06	1.81E-04	1.8
	1000	2.30E-05	3.56E-04	1.2

Another metric that mission designers may care about is target access gap time. Fig. 13 shows how altitude affects the maximum, minimum, and average gap times. In general, the higher altitudes considered correspond to longer gap times. An exception occurs at the lowest altitude tested, where both the average and maximum gap times are larger than those at nearby altitudes. This is consistent with the lowest altitude being the sparsest of the altitudes compared. It should be noted that the maximum gap time, 1458 min, is not included for this altitude for the sake of plot readability.

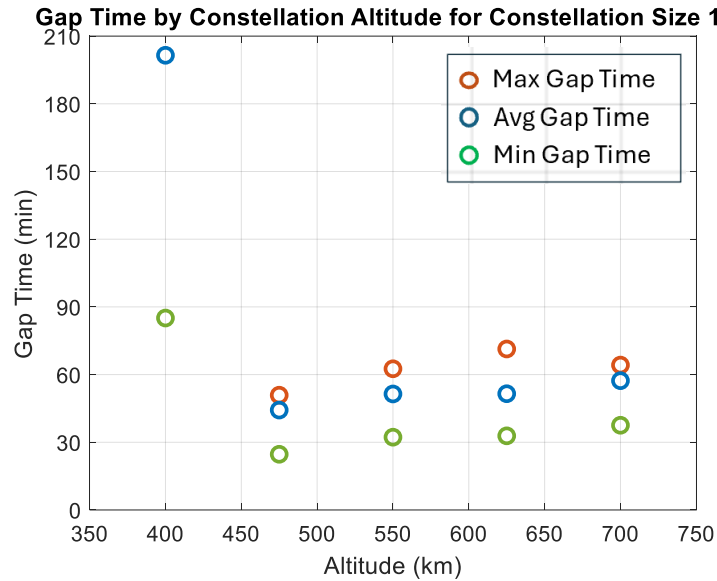


Fig. 13. Gap time by constellation altitude for constellation size 100.

This behavior provides useful insight for mission planning, especially when weighing trade-offs between altitude and coverage. This specific metric was not optimized in this process, but this capability could be added if that was a priority for a specific mission.

5.2 Earth Land Coverage: Preliminary

Another natural application of this method is the design of constellations that maximize Earth land coverage while also minimizing constellation collision risk. Earth land coverage is becoming increasingly important as the demand for LEO services continues to grow, particularly in areas such as broadband communications and internet access (e.g., Starlink, OneWeb, Kuiper) [1][20], Earth observation and remote sensing [21], and weather and climate monitoring [22]. As these constellations scale, the need for optimization methods that balance coverage and conjunction risk will only become more critical.

The same orbit constellation shells as described in Table 2 are used for this mission application. Also, the edge weights as given in Eq. 1 are used, with the coverage metric applicable to coverage of land area. Risk versus coverage results are presented in Fig. 14 below.

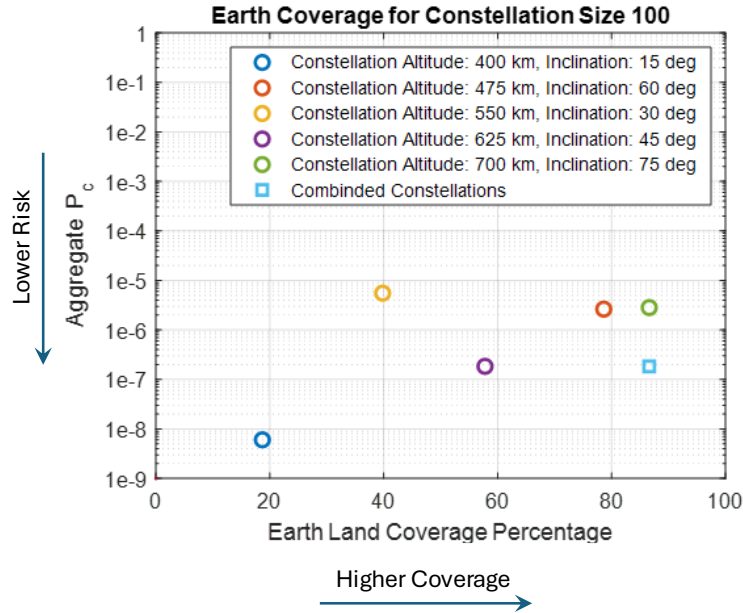


Fig. 14. Risk and Earth land coverage by constellation altitude.

It was observed that the coverage results for all three constellation sizes returned the same coverage results, meaning that there are redundant observations from the additional satellites, therefore only the constellation of size 100 results are shown. The algorithm reduced the aggregate P_c compared against the random selection of satellites presented in Table 5. The best coverage performance was obtained by constellations at higher inclinations. These pass over more land mass, which results in better coverage. The constellation drawn from all orbit shells produced the best results in both the risk and coverage metrics. This highlights the power of the algorithm to produce an optimized constellation design for designated metrics with improved results when the trade space of candidate satellite orbits is enlarged.

These results are preliminary and specific to the Earth land coverage example. The simulated annealing algorithm could be further tuned to improve performance for this particular scenario. The intent of this subsection is not to present a final solution, but rather to demonstrate the flexibility of the method in addressing additional mission objectives.

6. CONCLUSION

In this work, we proposed a graph-based optimization method for designing high-density LEO constellations within already dense LEO environments and demonstrated that this method can identify satellite distributions that minimize collision risk while maximizing mission-specific objectives. The goal was not to produce a single, fixed constellation, but rather to demonstrate a viable approach that can be applied to large, complex orbital environments.

The framework is flexible: additional objectives, such as global coverage, communication performance, or Earth observation priorities can be incorporated without changing the underlying optimization process. By representing satellites as nodes and potential interactions as weighted edges, this method allows for scalable evaluation and selection of satellite subsets that meet multiple mission objectives, providing a practical tool for planning safe and efficient LEO constellations.

Future work may focus on the practical implementation of these models through simulations, aiming to support policy development and international collaboration in space sustainability efforts. Additional future work may include running simulations over longer timespans. Lastly, the LEO object catalog may be augmented through the use of the NASA Orbital Debris Engineering Model (ORDEM) that has models for the debris population that are too small to be tracked by active sensors.

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